

PROBLEM SET 3

ECON301—ECONOMETRICS I

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Instructions: Due on 20/3/2021; scan or type your work and submit it through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. Let $\{W_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables with $E(W_i) = 0$ and $\text{Var}(W_i) = \sigma_W^2$, show that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i \rightarrow_d N(0, \sigma_W^2)$$

2. Let $\{Y_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables with Bernoulli probability mass functions with unknown parameter p . Let $\hat{p} = \frac{1}{n} \sum_{i=1}^n Y_i$.

- (a) Find and justify the large sample distribution of

$$\frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

- (b) Find and justify a consistent estimator of $\text{Var}(X_1)$.

- (c) Find and justify the large sample distribution of

$$\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

- (d) A sample of size 127 is such that $\sum_{i=1}^{127} Y_i = 39$. Find a 99% confidence interval for p .

3. Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of random variables such that $E(X_n) \rightarrow c$ and $\text{Var}(X_n) \rightarrow 0$ as $n \rightarrow \infty$. Show that then $X_n \rightarrow_P c$.

4. **Chebychev's second inequality.** Let X be a random variable with $E(X) = \mu < \infty$ and $\text{Var}(X) = \sigma^2$ with $0 < \sigma^2 < \infty$. Then for any real number $k > 0$,

$$\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Prove this result.

5. Suppose $\bar{X}$ is the mean of 100 observations from a population with mean μ and variance $\sigma^2 = 9$. Find limits between which $\bar{X} - \mu$ will lie with probability at least 0.90 using:

- (a) The second Chebychev's inequality from item 4.
(b) The Central Limit Theorem.

Comment on your results. Which limits are tighter?